

UTQIAGVIK, AK, USA

WMO: 700260

Lat: 71.283N

Lon: 156.782W

Elev: 10

StdP: 101.21

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 27502

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.5	-35.8	-37.1	0.1	-33.7	-36.1	0.1	-33.1	16.0	-19.5	14.6	-18.8	3.3	200	1.104

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	14.6	11.7	12.3	10.3	10.3	8.6	11.9	14.0	10.4	12.2	8.8	10.3	5.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	7.9	13.2	8.9	7.1	11.5	7.6	6.5	9.8	33.8	14.1	30.1	12.3	26.6	10.2	16.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.9	12.2	11.1	DB	-40.7	19.1	3.4	2.4	-43.2	20.9	-45.1	22.3	-47.0	23.7	-49.5	25.5	
			WB	-36.4	14.4	1.9	1.4	-37.7	15.4	-38.8	16.2	-39.8	17.0	-41.2	18.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.9	-23.9	-23.7	-23.5	-15.5	-5.2	2.1	5.4	4.4	1.2	-5.9	-14.4	-20.9
	DBStd	12.34	6.62	7.32	6.48	5.96	4.78	3.14	3.39	3.25	3.14	4.95	6.04	6.10
	HDD10.0	7279	1052	944	1038	766	473	237	148	177	263	492	731	959
	HDD18.3	10310	1310	1178	1296	1016	731	487	400	431	513	751	981	1217
	CDD10.0	10	0	0	0	0	0	1	6	4	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.7	5.7	6.0	5.4	5.4	5.5	5.4	5.5	5.5	5.6	6.3	6.2	5.8
Precipitation	PrecAvg	127	3	5	4	4	6	10	23	27	20	12	7	5
	PrecMax	214	15	21	23	18	22	36	51	66	47	37	21	29
	PrecMin	51	0	0	0	0	0	0	2	7	4	5	2	1
	PrecStd	36	4	5	5	4	5	8	13	15	11	8	4	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.9	-2.3	-2.6	1.5	4.5	14.9	18.3	17.1	12.6	3.3	0.6	-2.1
		MCWB	-2.5	-2.7	-2.9	0.7	3.3	11.0	13.5	13.1	9.8	2.4	-0.2	-2.6
	2%	DB	-6.4	-5.7	-6.6	-2.2	2.3	11.2	15.2	13.8	9.1	1.8	-1.2	-5.8
		MCWB	-6.9	-6.0	-6.8	-2.8	1.4	8.5	12.1	11.6	7.9	0.9	-2.0	-6.4
	5%	DB	-12.2	-9.4	-10.4	-4.6	1.3	8.8	12.8	11.6	7.2	0.8	-3.1	-8.8
		MCWB	-12.7	-9.8	-10.9	-5.1	0.6	6.9	10.6	10.2	6.4	-0.1	-3.9	-9.3
	10%	DB	-15.4	-13.4	-14.3	-7.1	0.6	6.7	10.8	9.5	5.3	-0.2	-5.1	-12.0
		MCWB	-15.7	-13.9	-14.7	-7.5	0.0	5.2	9.2	8.4	4.7	-0.9	-5.8	-12.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.2	-2.5	-2.8	0.7	3.3	11.6	14.3	13.8	10.1	2.7	-0.1	-2.5
		MCDB	-1.8	-2.3	-2.5	1.4	4.3	14.5	17.5	16.0	12.3	3.1	0.2	-2.2
	2%	WB	-6.4	-5.8	-6.8	-2.6	1.6	8.8	12.3	11.8	8.0	1.0	-2.0	-6.3
		MCDB	-5.9	-5.5	-6.6	-2.2	2.1	11.1	14.8	13.4	9.1	1.6	-1.4	-5.8
	5%	WB	-12.3	-9.5	-10.6	-5.1	0.8	7.0	10.7	10.2	6.5	0.1	-3.8	-9.3
		MCDB	-11.9	-9.1	-10.2	-4.6	1.2	8.8	12.7	11.7	7.2	0.7	-3.2	-8.9
	10%	WB	-15.4	-13.3	-14.5	-7.5	0.1	5.3	9.2	8.5	4.8	-0.9	-5.9	-12.4
		MCDB	-15.2	-12.8	-14.1	-7.2	0.5	6.7	10.8	9.5	5.3	-0.3	-5.3	-11.9
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.5	6.7	6.7	4.3	4.9	6.1	4.6	3.4	4.3	5.7	6.0
		MCDBR	10.2	10.9	9.8	7.2	4.3	9.7	10.4	9.1	6.7	4.1	7.1	8.7
		MCWBR	10.0	10.8	9.7	6.9	3.7	7.5	7.7	6.7	5.4	4.0	6.9	8.6
	5% WB	MCDBR	10.4	11.0	10.0	7.2	4.2	9.6	10.1	8.8	6.6	4.0	7.4	8.7
		MCWBR	10.3	10.9	9.9	7.0	3.7	7.5	7.8	6.7	5.4	3.9	7.2	8.5
Clear-Sky Solar Irradiance	taub	N/A	0.203	0.245	0.305	0.315	0.313	0.334	0.330	0.283	0.202	N/A	N/A	
	taud	N/A	1.973	2.156	2.053	2.135	2.314	2.422	2.519	2.555	2.134	N/A	N/A	
	Ebn at Noon	0	465	752	802	850	872	838	798	739	477	0	0	
	Edh at Noon	0	35	69	112	118	103	90	72	50	28	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.28	1.84	4.09	5.57	6.43	5.06	2.80	1.32	0.33	0.00	0.00	
	RadStd	0.00	0.03	0.11	0.27	0.45	0.33	0.56	0.28	0.11	0.04	0.00	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.42	+2.42	+1.38	N/A	N/A	N/A	-429	-430	N/A	N/A
(47)	Regional (3 neighbors)	+1.46	+2.68	+1.38	N/A	N/A	N/A	-483	-485	N/A	N/A

Nomenclature: See separate page